

the present. Survey of India has compiled a list of 157 alien invasive animal species. Introduction of water hyacinth in tropical countries rapidly multiplies in eutrophic rivers and lakes threatens the survival of many aquatic species.

Most ecologically sensitive regions of India, including the biodiversity hotspots, islands, coastal forests, freshwater swamp forests, mangroves and forest reserves, coincide with the identified 'invasion hotspots', indicating their vulnerability to alien plant invasion.

4.9. CONSERVATION OF BIODIVERSITY

It is the preservation of ecosystem, species and genetic diversity in a scientific manner so as to maintain it at its threshold level and derive sustainable benefits for the present and future generation. Over population and increased demand are greatly disturbing the natural resources and its diversity at all the levels. Humans derive a number of economic benefits from biodiversity of nature like food, wood, fibers, drugs medicines etc. Biodiversity plays an important role in maintaining and sustaining the supply of goods and services. Industrialization and over exploitation of resources becoming a great threat to the wild life. This often leads to the extinction of species. So it is also important to preserve every species of plants, animals and even microorganisms as each one has some or the other role to play in our nature. The most effective and efficient mechanism for conserving biodiversity is to prevent further destruction or degradation of habitats. To attain this, two basic strategies developed are (i) in-situ conservation and (ii) ex-situ conservation.

4.9.1. IN-SITU CONSERVATION STRATEGIES

It refers to the conservation of biological diversity within their natural habitats. It is being done by declaring the area as "protected area", with emphasis either to save the entire area or an endangered species.

Protected Areas

These are areas of land and/or sea especially dedicated to the protection and maintenance of biological diversity, and of natural and associated cultural resources. The protected areas are managed through legal or other effective means. There are several areas throughout the world which protect rare species or wilderness. In Indian context we have biosphere reserves, national parks

and sanctuaries. Apart from that various projects and programmes are also developed to preserve endangered species. These include : (i) project tiger (ii) project elephant (iii) crocodile conservation programmes etc.

1. Biosphere reserves

The Biosphere Reserves are a special category of protected areas of land or coastal environments, wherein people are an integral component of the system. Biosphere Reserves provides support for research monitoring, education and information exchange related to local, national and global issues of conservation and development.

A Biosphere Reserve consists of three zones—Core, buffer and transition zones.

(i) **Core Zone** : It comprises an undistributed and legally protected ecosystem. No human activity is allowed in this zone.

(ii) **Buffer Zone** : It surrounds the core area. Buffer zones are open for research, education, tourism, sustainable agriculture, beneficial and ecologically sound development programmes.

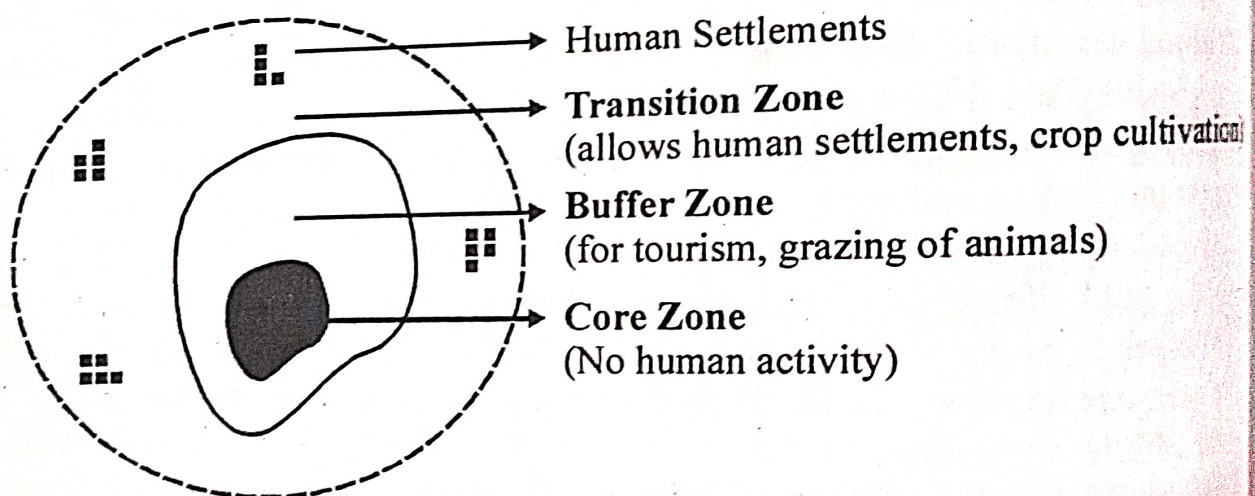


Fig. 2. Core, buffer, and transition area zonation in biosphere reserve.

(iii) **Transition Zone** : It is the outer most part of the Biosphere Reserve. It serves as an area of active cooperation between reserve management and the local people, wherein activities like settlements, cropping, forestry, recreation and other economic uses continue in harmony with conservation goals.

The Indian government has established 18 Biosphere Reserves of India. Three important biosphere reserves listed below :

(a) **Nilgiri Biosphere Reserve** : Western Ghats and encompasses parts of Tamilnadu, Kerala and Karnataka.

(b) **Gulf of Mannar** : Tamilnadu, south eastern tip of India.

(c) **Sundarbans** : West Bengal

2. National Parks and Sanctuaries

The national park provides protection to the entire set of the ecosystem, i.e., flora, fauna, landscape, etc., of that region where as wildlife sanctuaries refer to an area which provides protection to the wild animals. Human interferences and tourism activities are not allowed in national parks where as these activities are allowed to a certain extent in sanctuary area. Our country has good network of national parks and sanctuaries. Most of the Sanctuaries are present inside the biosphere reserves or national parks.

(i) **National Parks** : A national park is an area which is strictly reserved for the welfare of wildlife where activities such as forestry, grazing or cultivation are not allowed. Private ownership right and habitat manipulation are not permitted in a National Park. At present, there are 104 National Parks in our country and some are listed below :

(a) **The Jim Corbett National Park, Uttarakhand** was the first National Park established in India.

(b) **Kaziranga National Park, Assam** is the only natural habitat of the endangered One-horned Rhinoceros in India. It is also famous as a Tiger Reserve.

(c) **Periyar National Park, Kerala** is nestled on the Western Ghats is a popular Project Tiger Reserve.

(d) **Bandipur National Park, Karnataka** is a part of the Nilgiri Biosphere Reserve.

(ii) **Sanctuaries** : Wildlife sanctuaries refer to an area which provides protection and favorable living conditions to the wild animals, birds etc. Operations such as harvesting of timber, collection of minor forest products and private ownership rights are allowed, which could be done without affecting the wild life. There are 492 wild life sanctuaries in India, some are listed below :

(a) **Gir Wildlife Sanctuary, Gujarat** is the sole home of Asiatic Lions in India.

(b) **Periyar Sanctuary, Kerala** is the protected area for elephants.

(c) **Manas Wildlife Sanctuary, Assam** is the protected area for the tiger, rare golden langur and the red panda.

(d) **Annamalai Sanctuary, Tamil Nadu** is the protected area for elephants and tiger.

(e) **Kumarakom Sanctuary, Kerala** is famous for birds.

(f) **Salim Ali Sanctuary, Goa** is famous for birds.

3. Project Tiger

This project was launched in 1973 to save the Tiger from extinction. At present there are 27 tiger reserves in our country. This project has helped to increase the number of tigers.

4. Project Elephant

This project was launched in 1991-1992 with the objective of saving the Asiatic elephant, and also to ensure long term survival of viable population of elephants.

5. Crocodile Conservation

This project was launched in 1975 to protect the remaining population of crocodile in their natural habitat. Crocodile have been threatened as their skin is used for making leather articles. This led to the near-extinction of crocodiles in the wild.

6. Sacred Forests and Sacred Lakes.

There has been traditional practice in India and some other Asian countries to maintain sacred forests and lakes to protect biodiversity. The sacred forests are protected by the tribal communities due to religious sanctity accorded to these forests.

Sacred groves of India are forest fragments of varying sizes, which are communally protected, and usually have a significant religious motive for the protecting community. Hunting and logging are usually strictly prohibited within these patches.

Indian sacred groves are often associated with temples, monasteries, shrines or with burial grounds. For example, Sacred tree groves in hinduism and Sacred deer parks in buddism. There are about 1000 square kilometres of undisturbed sacred groves, scattered in patches all over the country. Only the groves in the remote and inaccessible areas remain untouched.

The sacred forests represent islands of pristine forests *i.e.*, most undisturbed forests without any human impact. These forests have been free from all disturbances, though these are surrounded by highly degraded landscapes. Many states in our country such as Karnataka, Maharashtra, Kerala, Meghalaya etc., possess sacred forests. These are serving as protective centers for a number of rare, endangered and endemic taxa. Similarly, aquatic flora and fauna is also protected in sacred water bodies. For instance Khecheopalri Lake in Sikkim has been declared sacred by the people to save aquatic life from degradation.

4.9.2. EX-SITU CONSERVATION STRATEGIES

It refers to the conservation of species in suitable locations away from their natural habitat. The need to conserve species in ex-situ arise when its population is so fragile/fragmented that its survival in wild may no longer be possible. These threatened or endangered species needs special considerations in food and shelter. It is necessary to create favourable environmental conditions for its survival. For ex- situ conservation of biodiversity, germ plasm banks or genre banks are established. These include botanical gardens, zoos, genetic resources centres, pollen grains, seed, and seedling, tissue culture and DNA banks. Some of the ex-situ conservation methods include the following :

- (i) **Botanical gardens** : Collections of special types of herbs, shrubs plants, threatened and endangers species are specially conserved here.
- (ii) **Zoological gardens** : Wild animals are giving favourable environment for their conservation.
- (iii) **Aquariums** : Provide better environment for aquatic plants or animals. Cultivation of fishes, crustaceans, amphibians etc., can be done in aquariums.
- (iv) **Research centres** : Do research activities for the conservation and preservation of organisms.
- (v) **Gene banks** : Field gene banks are other ways of preservation of genetic resources under normal growing conditions. Genes of threatened/ endangered species can be preserved.
- (vi) **Seed banks** : Seed gene banks are the easiest way to store germ plasm of wild and cultivated plants at low temperature in cold rooms.
- (vii) **In-vitro cultivation techniques (in glass)** : Tissue culture methods are now widely used to cultivate organisms in suitable growth media.
- (viii) **Cryo preservation** is the use of very low temperatures to preserve structurally intact living cells and tissues. Ultra low temperature (*i.e.* at a temperature of -196°C in liquid nitrogen) is useful for conserving seeds, vegetative propagated parts, tissue culture, etc.